FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ (SOLUTIONS)

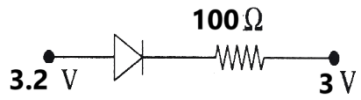
PHYSICS

CLASS XII

SEMICONDUCTOR ELECTRONICS

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1. The current in the circuit shown in the figure considering an ideal diode is



(a) 20 A (b) 2 x 10^{-3} A (c) 200 A (d) 2 x 10^{-4} A

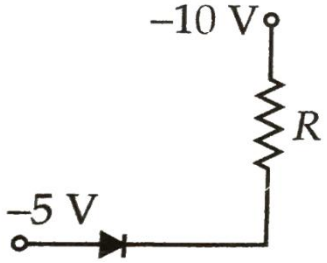
Ans: (b) 2 x 10^{-3} A

Potential difference across diode $= 3.2 - 3 = 0.2\,V$

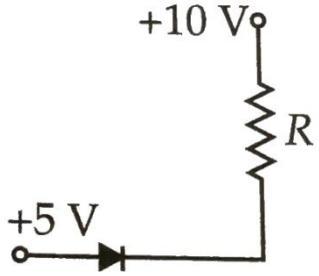
$\therefore$ Current through diode $i = \frac{0.2}{100} = 2 \times 10^{-3} A$

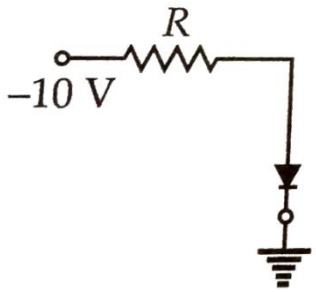
2. When a junction diode is reverse biased, the flow of current across the junction is mainly due to
   (a) diffusion of charges (b) the nature of material
   (c) drift of charges (d) both drift and diffusion of charges
   Ans: (c) drift of charges
3. A donor impurity results in
   (a) production of n- semiconductor (b) production of p- semiconductor
   (c) increase of resistance of the semiconductor (d) energy bands just above the filled valency
   Ans: (a) production of n- semiconductor
4. Which of the following p-n junction is forward biased?

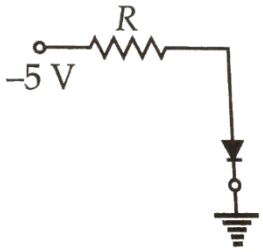
(a)



(b)



(c) 

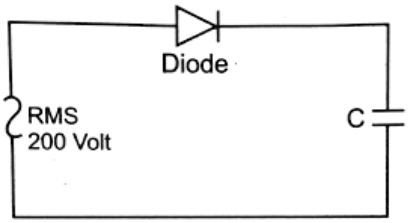
(d) 

Ans: (a)

A diode is said to be forward biased when the P-type side (anode) is connected to a higher potential (positive terminal) and the N-type side (cathode) is connected to a lower potential (negative terminal).

5. If no external voltage is applied across a p-n junction, there would be
   (a) no electric field across the junction
   (b) an electric field pointing from n- type to p- type side across the junction
   (c) an electric field pointing from p- type to n- type side across the junction
   (d) a temporary electric field during the formation of p-n junction that would subsequently disappear
   Ans: (b) an electric field pointing from n- type to p- type side across the junction

6. In the circuit given in the figure, an a.c source of 200 V is connected through a diode D to a capacitor. The potential difference across the capacitor will be

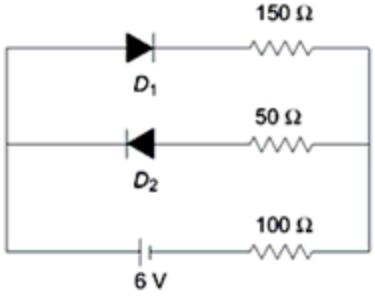


   (a) 100 V (b) 283 V (c) 200 V (d) 310 V
   Ans: (b) 283 V
   A diode conducts only during the positive half cycle of a.c. Accordingly the capacitor charges and discharges. During charging,

$$\text{Voltage across capacitor} = E_{rms}\sqrt{2} = 200 \times \sqrt{2}V = 282.8V \approx 283V$$

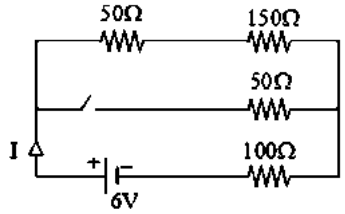
7. If a full wave rectifier circuit is operating from 50 Hz mains, the fundamental frequency in the ripple will be
   (a) 50 Hz (b) 70.7 Hz (c) 100 Hz (d) 25 Hz
   Ans: (c) 100 Hz

8. The circuit shown in the figure contains two diodes each with a forward resistance of 50 Ω and with infinite backward resistance. If the battery voltage is 6 V, the current through the 100 Ω resistance is



(a) 0 A (b) 0.02 A (c) 0.03 A (d) 0.036 A

Ans: (b) 0.02 A



$I = \frac{V_{net}}{R_{net}}$ $I = \frac{6}{300}$

$= 0.02$ A

9. A Ge specimen is doped with Al. The concentration of acceptor atoms is $\approx 10^{21}$ atoms m^{-3}. Given that the intrinsic concentration of electron -hole pair is $\approx 10^{19}$ m^{-3}, the concentration of electrons in the specimen is

(a) 10^{17} m^{-3} (b) 10^{15} m^{-3} (c) 10^{4} m^{-3} (d) 10^{2} m^{-3}

Ans: (a) 10^{17} m^{-3}

$n_i^2 = n_e \times n_h$

where n_i is concentration of electron-hole pair and n_h is concentration of acceptor or holes.

Given, $n_i = 10^{19}$ per m^3, $n_h = 10^{21}$ per m^3

$(10^{19})^2 = n_e \times 10^{21}$

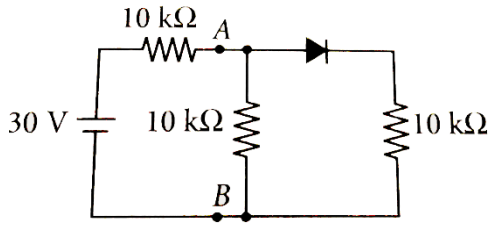
$\Rightarrow n_e = \frac{10^{38}}{10^{21}} = 10^{17}$ per m^3

10. The potential barrier in the depletion layer is due to

(a) ions (b) holes (c) electrons (d) forbidden band

Ans: (a) ions

11. In the given circuit, the potential difference between A and B is



(a) 0 V (b) 5 V (c) 10 V (d) 15 V

Ans: (c) 10 V

Here p-n junction is forward biased. If p-n junction is ideal, its resistance is zero. The effective resistance across A and B

$$= \frac{10 \times 10}{10 + 10} = 5K\Omega$$

Current in the circuit

$$I = \frac{30}{(10 + 5) \times 10^3} = \frac{2}{10^3}$$

$$\therefore \quad \text{Current in each arm} = \frac{I}{2} = \frac{1}{10^3} A.$$

Pot. diff. across A and B

$$= \frac{1}{10^3} \times 10 \times 10^3 = 10V$$

12. In semiconductors at room temperature
    (a) the valence band is partially empty and the conduction band is partially filled
    (b) the valence band is completely filled and the conduction band is partially filled
    (c) the valence band is completely filled
    (d) the conduction band is completely empty
    Ans: (a) the valence band is partially empty and the conduction band is partially filled
13. Reverse bias applied to a junction diode
    (a) lowers the potential barrier
    (b) raises the potential barrier
    (c) increases the majority carrier current
    (d) increases the minority carrier current
    Ans: (b) raises the potential barrier

For questions 14 to 18, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

14. **Assertion:** A pure semiconductor has negative temperature coefficient of resistance.
    Reason: On raising the temperature, more charge carriers are released, conductance increases and resistance decreases.
    Ans: A
15. **Assertion:** An n-type semiconductor has a large number of electrons but still it is electrically neutral.
    Reason: An n-type semiconductor is obtained by doping an intrinsic semiconductor with a pentavalent impurity.
    Ans: B
    A n-type semiconductor is formed by doping pure germanium or silicon crystal with suitable impurity atoms of valency five. As the impurity atoms take the position of Ge atom in germanium crystal, its four electrons form covalent bonds by sharing electrons with the neighbouring four atoms of germanium whereas the fifth electron is left free. Since the atom on the whole is electrically neutral, the n-type semiconductor is also neutral.

16. **Assertion:** The drift current in a p-n junction is from the n-side to the p-side.
    Reason: It is due to free electrons only.
    Ans: C
17. **Assertion:** The dominant mechanism for motion of charge carriers in forward and reverse biased silicon p- n junction are drift in both forward and reverse bias.
    Reason: In reverse biasing, no current flow through the junction.
    Ans: D
18. **Assertion:** The number of electrons in a P-type silicon semiconductor is less than the number of electrons in a pure silicon semiconductor at room temperature.
    Reason: It is due to law of mass action.
    Ans: A

- The law of mass action in semiconductors states that the product of the concentration of electrons (n) and holes (p) is constant at a given temperature, which can be expressed as:

$n_i^2 = n \cdot p$

- In a p-type semiconductor, since the concentration of holes (p) is increased due to doping, the concentration of electrons (n) must decrease to maintain this product constant. Thus, the law of mass action supports the assertion.